

These metals are also called the *d-block elements*.

H																		He
Li	Be											B	C	N	O	F	Ne	
Na	Mg											Al	Si	P	S	Cl	Ar	
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr	
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe	
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn	
Fr	Ra	Ac	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og	
		La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu		
		Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr		

Laboratory sample of pure titanium



Transition Metals

With 38 members, the transition metals form the largest set of elements in the periodic table. They include those that sit in periods (rows) 4 to 6 and groups (columns) 3 to 12. Some characteristics do vary between members of this block of elements, but they are mostly similar. For example, all these metals are typically hard and shiny, have high melting and boiling points, and are good conductors of heat and electricity.

Titanium

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Ti

Named after the titans of Greek mythology—gods who were known for their exceptional strength—titanium is hard, strong, but also lightweight. It's a fitting name because most of the metal's applications are based on titanium having the highest strength-to-weight ratio of all metals, as well as its ability to resist corrosion.

Everyday use

Titanium is used to make items where strength is important, but the weight of the object must be kept low. Despite its higher cost, this metal has replaced steel and aluminum in some everyday items, such as eyeglass frames.

Titanium is also useful in alloys—for example to make stronger steel.

Lightweight titanium frame is durable and resistant to impact



Glasses with titanium frames

Shine fades to gray when element is exposed to air

Artificial arm made of titanium and carbon fiber



Artificial parts

This strong and light element is used in the medical industry to repair and even replace damaged or lost bones, joints, and limbs. Titanium plates, screws, and rods can also hold bones together as they heal. Titanium's ability to resist corrosion is also very important in medical applications.